

Shubham Vij

Manager, Staff Research Engineer at Snap Inc.

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SUMMARY

Research Engineering manager and Tech Lead with 8+ years building large-scale Machine Learning systems. Spanning research productionization, distributed training/inference, and engineering team leadership for the following areas: Graph Neural Networks (GNNs), embedding-based retrieval, generative recommendation, and ML platform / MLOps infrastructure.

EXPERIENCE

Snap Inc

Manager, Staff Research Engineer

Apr 2024 - Present

Research Engineering Manager leading the Research Enablement Team @ Snap - accelerating the productionization of research across Graph ML and Generative Recommendation. Manage a team of research engineers, owning the technical roadmap and partnering cross-functionally to engineer scalable Machine Learning systems and ML infrastructure that realize value in large-scale recommendation and personalization systems. Team's scope includes applying internal and industry ML research - spanning Graph ML, LLMs, foundation models, transformers, and generative / sequential recommendation. Operate as Tech Lead across the broader User Modeling and Personalization Research org, driving ML platform and MLOps strategy.

Tech Lead, Senior Research Engineer

Aug 2021 - Apr 2024

Research Engineer

Feb 2018 - Aug 2021

Engineered ML systems to productionize Snap Research, building solutions for news content de-duplication and large-scale Graph Neural Network training — open-sourcing GiGL (Gigantic Graph Learning), Snap's distributed graph ML library. Contributions and tech-leading large scale Graph ML efforts through GiGL lead to various successful embedding-based retrieval and representation learning launches across Snap, powering friend recommendation, Search, Trust and Safety, Content, and ADS.

Engineered key features for Snap Publisher – a full-stack web tool for advertisers/publishers to post on Snapchat. Built framework and tooling, including full stack development, browser optimization, Computer Vision (CV) and AI features, to ease development of high-performance video composition and editing tools on the web.

Built and maintained the Research team's ML platform and cloud compute infrastructure, used by 100s of ML projects and practitioners to train models at scale via distributed training on GPU / TPU. Engineered MLOps tooling for cost optimization, scaling, and reporting. Helped engineer the foundations of projects across NLP and Computer Vision (CV) domains.

Software Engineer Intern

Jan 2017 - Apr. 2017

Designed a library for on-device (edge ML) offline keyword detection, later used as the technology backing various speech recognition lenses.

Amazon Inc

Software Development Engineer Intern

May 2016 - Aug 2016

Developed software to automate localization for the Amazon Silk browser, saving roughly 1+ hour/week of manual engineering effort.

IBM

Mobile Solutions Developer

Sep 2015 - Dec 2015

Consulted for one of the top 5 financial institutions in Canada, engineering and presenting business and technical solutions centred around client authentication to C-level executives.

Microsoft

Xbox Marketing Assistant

Jan 2014 - Apr 2014

Automated computing viability of rebate programs related to video game inventories. Designed and led a digital marketing campaign for the Canadian market for new AAA game launch - Titanfall.

EDUCATION

University of Waterloo

Bachelors in Computer Science

Waterloo, ON

Dec. 2019

Wilfrid Laurier University

Bachelors in Business Administration

Waterloo, ON

Dec. 2019

SKILLS SUMMARY

Languages: Python, Scala, Java, JavaScript, Objective-C, SQL, HTML, CSS
ML / AI: PyTorch, PyG, TensorFlow, NumPy, Transformers, LLMs, Generative AI
ML Systems: Kubernetes, Docker, Kubeflow, Bazel, distributed training, model serving
Big Data: Apache Beam, Spark, BigQuery
Web/Mobile: React, Redux, Node, Express, iOS
Cloud: GCP (Vertex AI), AWS

PATENTS

Cloud Based Machine Learning

US Patent 12,223,402

E Buehl, J Hurwitz, S Tulyakov, **S Vij**

Image Segmentation System

US Patent 10,964,023

S Vij, M Chai, D Lemieux, I Wehrman

Device Control Using Audio Data

US Patent 11,487,501

X Chen, Y Monastyrshyn, F Poliakov, **S Vij**

Messaging System with Trend Analysis of Content

US Patent 11,948,558

LRM das Neves, VS Sousa, **S Vij**

PUBLICATIONS

Beyond Unimodal Perspectives: Generative Retrieval with Multimodal Semantics

SIGIR '26, pp. 4414–4418

2026

J Zhu, M Ju, Y Liu, **S Vij**, D Koutra, N Shah, T Zhao

Gigl: Large-scale Graph Neural Networks at Snapchat

KDD '25, pp. 5225–5236

2025

T Zhao, Y Liu, M Kolodner, K Montemayor, ..., **S Vij**, N Shah

Training Industry-scale GNNs with GiGL

KDD '25, pp. 6241–6242

2025

Y Liu, T Zhao, M Kolodner, K Montemayor, **S Vij**, N Shah

Embedding Based Retrieval in Friend Recommendation

SIGIR '23, pp. 3330–3334

2023

J Shi, V Chaurasiya, Y Liu, **S Vij**, Y Wu, et al.